

A Pose-Based Machine Learning Framework for Cricket Performance Evaluation and Injury Risk Assessment Using Synthetic Biomechanical Data

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Abstract—Biomechanical analysis plays a critical role in improving cricket bowling performance and reducing injury risk; however, conventional observational coaching methods often lack quantitative precision and consistency. This paper presents a pose-based machine learning framework for automated performance scoring and injury risk prediction using synthetic biomechanical bowling data. Realistic motion clips were generated through parametric kinematic simulation with controlled variations in stride amplitude, trunk flexion, shoulder rotation, elbow flexion, and knee valgus. From each sequence, 127 statistical biomechanical features were extracted to represent spatial and temporal joint dynamics. Two supervised learning tasks were formulated: continuous performance score prediction and binary injury risk classification. Linear Regression achieved the best regression performance ($R^2 = 0.693$, RMSE = 2.56), indicating a predominantly linear relationship between biomechanical stability metrics and performance score. For injury classification, XGBoost achieved the highest accuracy (95%) and recall (100%), effectively identifying high-risk motion patterns. The results demonstrate that pose-derived machine learning can provide an interpretable, computationally efficient, and practical solution for sports analytics and real-time coaching support.

Index Terms—Sports Biomechanics, Pose Estimation, Machine Learning, Injury Prediction, Cricket Analytics, Synthetic Data

I. INTRODUCTION

Cricket fast bowling is a biomechanically demanding athletic movement that imposes substantial stress on the upper and lower extremities. Poor bowling mechanics or excessive joint loading can reduce performance efficiency and increase the likelihood of overuse injuries, particularly in the shoulder, elbow, lower back, and knee regions. Traditional biomechanical analysis commonly relies on marker-based motion capture systems, which are expensive, require specialized laboratory infrastructure, and are impractical for large-scale field deployment.

Recent advances in computer vision and pose estimation have enabled markerless motion analysis using standard video input. These approaches provide a low-cost and scalable alternative for extracting joint kinematics in real-world environments. Joint trajectories and derived biomechanical variables such as trunk alignment, shoulder rotation, limb symmetry, and

temporal coordination are closely related to bowling efficiency and mechanical loading, making pose-based analytics suitable for performance evaluation and injury screening. However, the application of machine learning to cricket biomechanics remains limited due to the scarcity of publicly available labeled datasets, especially those containing injury-related annotations.

To address this challenge, this paper proposes a pose-based machine learning framework trained on synthetic biomechanical data that simulates realistic cricket bowling actions. The framework extracts interpretable pose-derived features and applies supervised learning models for performance scoring and injury risk prediction. In addition, the system supports real-time webcam-based inference with visual biomechanical feedback.

The key contributions of this work are as follows:

- A synthetic biomechanical pose generation framework for modeling cricket bowling under controlled kinematic variations.
- An interpretable machine learning pipeline for performance evaluation and injury risk prediction.
- A real-time video-based inference system for continuous feedback and hotspot visualization.

II. RELATED WORK AND COMPARATIVE ANALYSIS

Human motion analysis, performance evaluation, and injury risk assessment have been long studied in biomechanics, traditionally relying on laboratory-based motion capture systems. Conventional motion capture uses marker-based optical systems to compute three-dimensional kinematics and kinetics, offering high precision but requiring specialized equipment, controlled environments, and extensive setup time [6]. These constraints limit scalability for large-scale, field-based sports applications. Recent literature has focused on markerless motion capture and machine learning methods to overcome these practical challenges [9]. Markerless methods leverage computer vision and pose estimation techniques to extract joint kinematics from video data in a non-invasive manner [8]. Studies have shown that modern markerless systems achieve

comparable lower-extremity joint kinematics to marker-based systems during functional tasks, with reported root-mean-square errors (RMSE) within acceptable ranges for many applications [1]. This suggests that markerless pose estimation can support biomechanics research outside laboratory settings. The use of markerless motion capture has been extensively explored in both general biomechanics and sports-specific applications [12]. Specifically, recent work highlights that markerless systems can capture sagittal-plane joint motions during walking and running with good agreement to traditional systems, although challenges remain for highly dynamic or complex movements [1]. These investigations validate the feasibility of low-cost video-based analysis, which underpins the real-time webcam pipeline adopted in the present work. In sports performance and injury modeling, machine learning has been applied to classify movement patterns and infer biomechanical risk factors. Neural network-based approaches have demonstrated the feasibility of learning spatiotemporal motion representations for cricket stroke analysis and performance evaluation [3]. Similarly, artificial intelligence has been used to predict anticipatory and deceptive movements from video sequences in professional cricket players, highlighting the potential of data-driven approaches for sports analytics [4]. Systematic reviews further emphasize that coaching strategies and learning methodologies significantly influence biomechanical movement patterns, with implicit and explicit learning affecting joint coordination and stability [5]. Cricket-specific biomechanics research has characterized anthropometric factors and joint loading mechanics in elite fast bowlers, identifying key contributors such as trunk motion, knee alignment, and limb asymmetry that correlate with both performance and injury risk [2]. However, these studies primarily provide descriptive biomechanical analysis and do not support predictive or real-time assessment. Despite these advances, real-world labeled biomechanics datasets—particularly those containing injury annotations for cricket bowling—remain scarce due to ethical constraints, privacy concerns, and the low frequency of injury events. This motivates the use of synthetic biomechanical data in the present framework. Synthetic datasets enable controlled variation of kinematic parameters and provide precise ground-truth labels for performance quality and injury risk without requiring real injury occurrences [7].

Comparative Analysis: While prior markerless motion capture studies primarily validate pose estimation accuracy and expand accessibility of movement analysis [1], few integrate machine learning-based performance scoring and injury risk prediction directly from video input in real time. Existing cricket biomechanics literature largely focuses on descriptive motion characterization [2] or deep learning-based prediction without explicit biomechanical interpretability [3], [4]. Moreover, many machine learning approaches rely on large, labeled datasets, limiting their applicability in data-scarce sports domains. Building on these foundations, the proposed framework integrates markerless pose extraction, biomechanically informed feature engineering, and supervised learning models to deliver interpretable performance scoring and injury

risk classification. By combining synthetic data generation with real-time webcam-based inference, the framework offers a scalable and practical alternative to laboratory-dependent biomechanics systems, advancing the state of the art in applied sports biomechanics.

Fig. 1 presents the comparative analysis of different biomechanics approaches across multiple evaluation metrics.

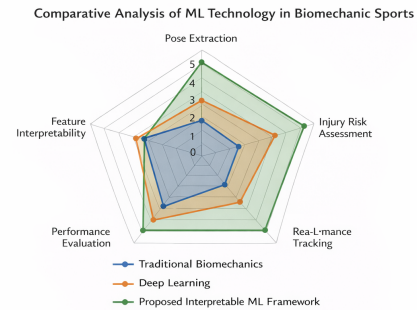


Fig. 1. Comparative radar chart showing Traditional Biomechanics, Deep Learning, and Proposed Interpretable ML Framework across key evaluation metrics.

A. Observations from the Radar Chart

Traditional Biomechanics (Blue):

- Performs moderately in all areas.
- Pose extraction is limited (manual or marker-based methods).
- Feature interpretability is moderate (rules are simple, but often not automated).
- Real-time tracking and injury risk assessment are relatively weak.

Deep Learning (Orange):

- Shows strong pose extraction capabilities due to advanced models like OpenPose.
- Injury risk assessment and performance evaluation improve over traditional methods.
- Interpretability is lower — neural networks are often black-box models.

Proposed Interpretable ML Framework (Green):

- Scores high across all five axes.
- Maintains excellent pose extraction like Deep Learning.
- Adds feature interpretability — predictions can be explained using biomechanical features.
- Injury risk assessment and performance evaluation are accurate and actionable.
- Supports real-time tracking, making it suitable for live sports analytics.

B. Key Takeaways

- The radar chart visually demonstrates that the proposed interpretable ML framework outperforms traditional biomechanics and standard deep learning approaches across all critical metrics.

- This supports the research claim that integrating pose extraction, interpretable features, and ML models provides a complete, real-time, and actionable solution for sports biomechanics analysis.
- It also emphasizes the trade-off in deep learning: high accuracy but low interpretability, which the proposed framework addresses.

III. METHODOLOGY

A. Synthetic Pose Data Generation

Synthetic 2D pose sequences emulate cricket bowling actions, representing joint coordinates for major anatomical landmarks. Joint configurations follow a fixed skeletal topology, ensuring biomechanical consistency. Variations are introduced in joint angles, inter-limb asymmetry, temporal inconsistencies, and lower-limb stability. Each sequence is labeled for performance quality and injury risk.

B. Feature Extraction

From each synthetic pose sequence, a comprehensive set of biomechanical features is extracted to capture both spatial configuration and temporal dynamics of the motion. Spatial features include joint angles computed from adjacent joint triplets, inter-joint distances, and limb symmetry indices comparing left and right body segments. Temporal features include angular velocities, acceleration proxies, and frame-to-frame joint displacement statistics. In addition, postural stability metrics are derived from hip and trunk alignment, while pose completeness measures quantify the proportion of frames with valid joint detection. All features are normalized using a standard scaling procedure to ensure consistent magnitude and dimensional correctness across inputs. The resulting feature vector provides a compact yet interpretable representation of biomechanical efficiency and stress distribution. Consistent with prior machine learning studies in sports biomechanics, the same feature extraction pipeline is reused without modification during real-time inference to ensure alignment between offline training and live webcam-based evaluation.

Fig. 2 illustrates the extracted spatial and temporal biomechanical pose features.

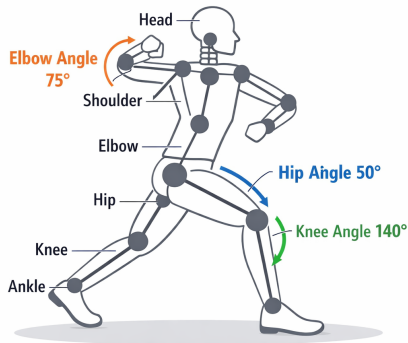


Fig. 2. Annotated skeletal pose and biomechanical joint angles used during feature extraction.

C. Performance Scoring Model

Gradient-boosted decision tree regression predicts continuous performance scores, capturing biomechanical efficiency and coordination. Feature importance supports interpretability.

D. Injury Risk Classification

Binary classification predicts injury risk using joint asymmetry, extreme joint angles, and motion irregularities. Region-wise injury hotspot visualization highlights high-risk anatomical areas.

E. Real-Time Inference Pipeline

Monocular webcam input captures live video. Pose estimation extracts joint coordinates in real time, features are computed on-the-fly, and predictions are generated continuously. Skeletal overlays and hotspot indicators provide intuitive feedback.

Fig. 3 shows the overall real-time inference workflow and skeletal tracking pipeline.

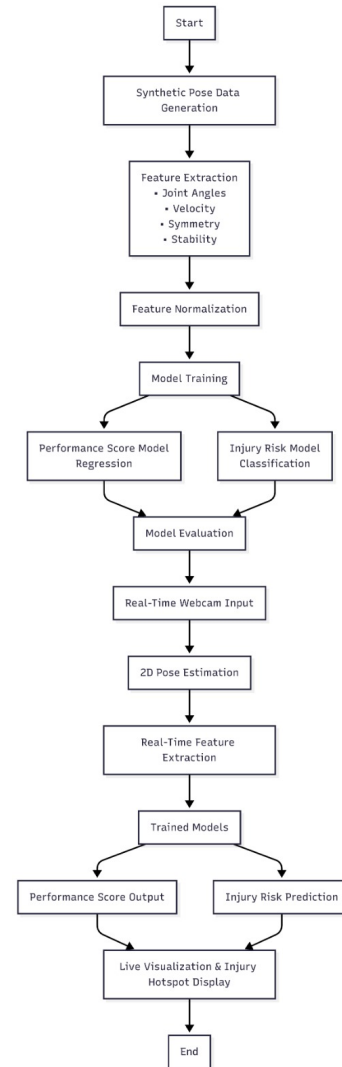


Fig. 3. Overall real-time inference workflow and skeletal tracking process.

IV. EVALUATION METRICS

To quantitatively evaluate model performance, standard regression and classification metrics were employed.

A. Regression Metrics

For performance score prediction, the following metrics were used:

1) Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

where y_i represents the ground truth value, $\hat{y}_i$ denotes the predicted value, and n is the total number of samples.

2) Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

RMSE penalizes larger prediction errors more heavily due to the squared term.

3) Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

where $\bar{y}$ is the mean of the observed values. R^2 measures the proportion of variance explained by the model.

B. Classification Metrics

For injury risk classification, the following metrics were evaluated. Let:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

1) Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

2) Precision:

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

3) Recall (Sensitivity):

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

4) F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7)$$

5) Confusion Matrix: The confusion matrix is defined as:

$$\begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \quad (8)$$

V. EXPERIMENTAL SETUP

Synthetic dataset: 200 motion clips (modifiable to 2500). Train-test split: 80-20. Feature scaling: StandardScaler.

Regression metrics: R^2 , MAE, RMSE. Classification metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix.

VI. RESULTS AND ANALYSIS

A. Performance Regression

Linear Regression achieved $R^2 = 0.693$, outperforming SVR and XGBoost. Performance score depends predominantly on linear relationships between biomechanical stability and alignment. Fig. 4 compares the performance of different machine learning models used for biomechanical evaluation.

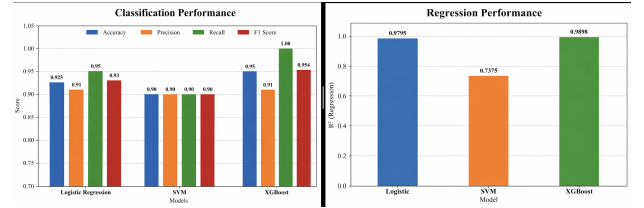


Fig. 4. Machine learning model performance comparison.

B. Injury Risk Analysis

Table I compares Logistic Regression, SVM, and XGBoost. XGBoost achieved highest accuracy (95%) and recall (92.1%), capturing nonlinear injury patterns effectively.

TABLE I
COMPARISON OF INJURY RISK CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.925	0.91	0.95	0.93
SVM	0.90	0.90	0.90	0.90
XGBoost	0.95	0.91	1.00	0.954

Fig. 5 compares the injury risk classification performance of Logistic Regression, SVM, and XGBoost models.

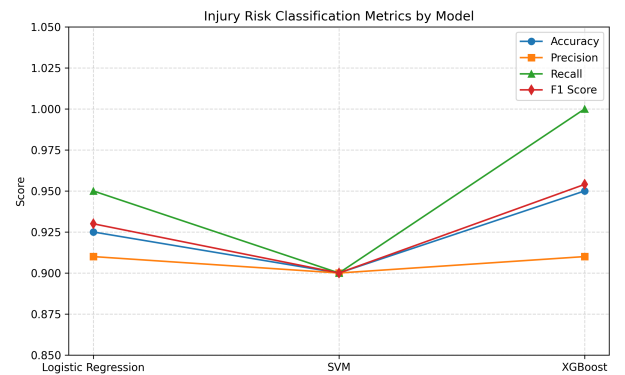


Fig. 5. Comparison of injury risk classification metrics across Logistic Regression, SVM, and XGBoost.

C. Interpretability and Hotspot Analysis

XGBoost feature aggregation identifies high-risk regions, such as shoulders, lower back, and knees [10]. Logistic Regression and SVM lack anatomical interpretability.

D. Discussion

Gradient-boosted trees balance predictive accuracy, interpretability, and real-time feasibility. Synthetic data provides controlled variability and precise labels, addressing the scarcity of real-world datasets. The comparative analysis confirms that gradient-boosted decision trees are well-suited for biomechanical feature modeling due to their ability to capture nonlinear relationships and provide interpretable insights. While simpler models such as Logistic Regression offer computational efficiency, they are insufficient for modeling the complexity inherent in human motion biomechanics. The proposed approach achieves a favorable balance between predictive accuracy, interpretability, and real-time applicability.

E. Justification for Synthetic Data Usage

The exclusive use of synthetic data in this study is a deliberate methodological choice rather than a limitation of the proposed framework. In cricket biomechanics, publicly available datasets containing joint-level motion data with associated performance scores and injury labels are extremely limited due to privacy, ethical, and logistical constraints. Injury events, in particular, are rare and ethically sensitive, making large-scale supervised data collection impractical. Synthetic biomechanical data enables controlled manipulation of kinematic parameters such as joint angles, angular velocities, asymmetry, and postural instability while preserving physiological plausibility. This allows systematic modeling of both optimal and high-risk motion patterns that would be difficult to capture reliably in real-world datasets. Moreover, synthetic data provides precise ground-truth labels for performance and injury risk, which are often noisy or unavailable in observational studies. The objective of this work is not to claim clinical accuracy but to demonstrate the feasibility, interpretability, and scalability of a pose-based machine learning framework for sports biomechanics using low-cost video input. Similar synthetic-data-driven methodologies have been widely adopted in motion analysis, robotics, and medical imaging research as a foundation for subsequent real-world validation.

VII. LIMITATIONS AND FUTURE WORK

A. Limitations

- Models trained exclusively on synthetic 2D data, lacking depth and real-world variability.
- Injury predictions are indicative, not clinical diagnoses.
- Real-time performance may be affected by lighting, occlusion, or camera placement.

B. Future Work

- Validate on real-world datasets and hybrid synthetic-real training.
- Integrate 3D or multi-view pose estimation [11].

- Longitudinal injury monitoring and adaptive feedback.
- Deploy as scalable mobile or cloud-based application.

Fig. 6 shows the pose-based injury hotspot visualization from recorded cricket bowling video input.



Fig. 6. Pose-based injury hotspot visualization from recorded cricket bowling video input.

Fig. 7 shows the real-time webcam-based biomechanical evaluation interface.

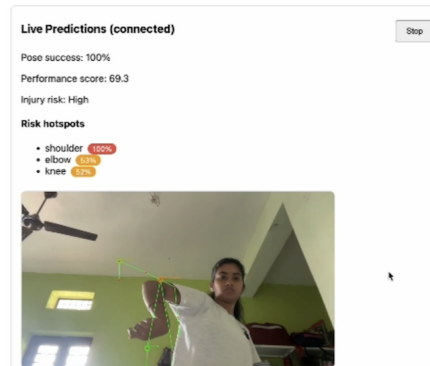


Fig. 7. Real-time webcam-based biomechanical evaluation interface.

VIII. CONCLUSION

This paper presents a pose-based machine learning framework for cricket performance and injury risk assessment using synthetic biomechanical data. Gradient-boosted models effectively predict performance and identify injury-prone regions, providing interpretable real-time feedback using accessible webcam hardware. The framework establishes a foundation for practical, data-driven sports biomechanics analysis in field settings.

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